

This is revision **65** to **Handbook Vol. 1 , CSP A-022-080.** **Quick Reference**

To bring this manual up to date, remove old pages and insert revised pages as follows:

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Model CL-600-2B19

QUICK REFERENCE HANDBOOK

Volume 1

CSP A-022-080

This handbook contains suggested procedures only for quick reference. These procedures in no way supersede current procedures outlined in the approved Airplane Flight Manual and any revisions thereto. In case of conflict, the Airplane Flight Manual takes precedence.

BOMBARDIER INC.
BOMBARDIER AEROSPACE COMMERCIAL AIRCRAFT
CUSTOMER SUPPORT

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REV 59	Introduces changes to the QRH to reflect Airplane Flight Manual Revision 59. • Harmonization to the other programs, Manual Changes Request (MCRs) and incorporation of all previous Temporary Revisions (TRs). • Relocation of historical data for Record of Revisions (i.e. covering REV 1 thru REV 58) and Record of Temporary Revisions (i.e. covering QRH TR RJ/1 thru TR RJ/167) to the Bombardier Aerospace online website.	BCSG	21 Mar 2012
REV 60	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 60.	BCSG	30 Nov 2012
REV 61	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 62.	BCSG	01 Oct 2014
REV 62	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 63.	BCSG	13 Feb 2015
REV 63	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 65.	BCSG	11 Jan 2016
REV 64	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 66.	BCSG	22 Apr 2016

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REV 65	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 67.		

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Record the date you insert each temporary revision in your manual.

Revision 59 is a major revision that incorporates all outstanding temporary revisions. A complete listing of all issued temporary revisions may be obtained from Bombardier Aerospace Customer Service.

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AIRPLANE OPTION CODES

Relevant pages of the QRH contain the following airplane option codes adjacent to the applicable text.

In cases where two or more option codes apply to the same text, and one of the options is the basic configuration, the option code <Type Spec> will appear, paired with other applicable option code(s). Example: <Type Spec> and <0004>.

Absence of an option code means that the data are applicable to all.

Airplane Option Codes		
Option Code	Customer Request	Description
<JAA>	CR 00-211	Data applicable to Joint Aviation Authorities registered airplanes only.
<0001>	CR 28-320	Metric Fuel Indication
<0002>	CR 00-008	Performance – Extended Range: 51000 lb MTOW
<0004>	CR 00-014	Performance – Long Range: 23995 kg MTOW
<0005>	CR 72-302	Power Plant - CRJ200 (CF34-3B1 Engine Basic Performance) <FAA> coded pages comply with FAR 25-109 at Amendment 25-41.
<0006>	CR 00-312	Performance – Additional 8-degree Flap setting
<0008>	CR 23-330	VHF Communication – Third Radio, 25 kHz Channel Spacing
<0010>	CR 23-350	Selcal
<0011>	CR 23-340	Single HF Radio
<0012>	CR 23-332	VHF Communication – Dual, 8.33 kHz Channel Spacing
<0013>	CR 25-359	Cargo Compartment – Automatic Temperature Control (40 Minute Diversion Capability)
<0014>	CR 23-334	VHF Communication – Third Radio, 8.33 kHz Channel Spacing
<0019>	CR 31-310	Data Concentrator Unit – Third DCU
<0020>	CR 33-310	Lights – Tail Logo Floodlights
<0021>	CR 33-303	Lights – Red Beacon Lights

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Airplane Option Codes		
Option Code	Customer Request	Description
<0024>	CR 34-332 CR 34-334 CR 34-336	Flight Management System (FMS) – Dual Installation
<0025>	CR 34-320	Inertial Reference System (IRS) – Dual Installation
<0026>	CR 34-380	Head-up Guidance System (HGS) – Category IIIa Capability
<0027>	CR 34-352	Global Positioning System (GPS) – Dual Sensors
<0039>	CR 31-320	EICAS 2000 Avionics Systems Update
<0040>	CR 34-328	Enhanced Ground Proximity Warning System (EGPWS)
<0045>	CR 34-360	Altimeter – Second Radio Altimeter
<0055>	CR 24-003	Electrical Power – Alternate Source for Cargo, Boarding and Maintenance Lights
<0071>	CR 35-006	Oxygen – Passenger, 22-Minute Chemical Oxygen Generators
<0096>	CR 34-312	Transponder – Mode S, Elementary Surveillance

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GENERAL NOTES, CAUTIONS AND WARNINGS

NOTE

- 1. Accomplish all the normal checklists up to and including the “Cleared to start” checklist before performing the following supplementary procedures for engine start.
- 2. If an external source is required, it should be capable of supplying 45 psi manifold pressure at the initiation of start, however, lower pressure units, dependent upon altitude and flow, have been proven satisfactory.

ENGINE START – EXTERNAL AIR

- (1) APU LCV Close
- (2) EngineSTART

———— **END** ————

ENGINE START – CROSS BLEED

- (1) APU LCV Close
- (2) 10TH STAGE bleed valve (on operating engine)Open
- (3) ECS pageSelect and monitor



Beware of increased jet blast when increasing thrust.

- (4) Thrust lever (for operating engine)Advance as required to achieve a minimum of 60 psi bleed air pressure (approximately 85% N₂).
- (5) L or R ENGSTART

NOTE

If thrust lever was advanced to achieve 60 psi, reduce to idle after starter cutout.

———— **END** ————

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ENGINE START – BATTERY / EXTERNAL AIR

- (1) FUEL, L BOOST PUMP ON
- (2) Ignition BARM
- (3) L ENG STARTSTART

———— **END** ————

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UNPRESSURIZED TAKE-OFF AND LANDING PROCEDURE

The procedures outlined below are recommended when performing a no-engine bleed take-off and landing, with the APU inoperative or selected off for performance reasons.

When the engine bleeds cannot be used for take-off (due to hot ambient temperature – short runway combination, etc.) and the APU is inoperative, the following procedures are recommended:

Before Take-off

- (1) PRESS CONT switchAUTO
- (2) EMER DEPRESS ON
- (3) L and R PACK switches OFF
- (4) 10TH STAGE, ISOL Closed
- (5) 10TH STAGE, L and R BLEED
AIRCLOSED
- (6) RAM AIR switch OPEN

———— **END** ————

At a safe altitude, preferably between 1500 and 3000 feet above field elevation:

After Take-off

- (1) RAM AIR switch Close
- (2) 10TH STAGE, L and R BLEED
AIR OPEN
- (3) L and R PACK switchesOn
- (4) EMER DEPRESSOff

NOTE

If an engine failure occurs, delay turning on the bleeds until after obstacle clearance is assured.

———— **END** ————

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Landing

The following procedures are recommended if engine bleeds are not to be used during landing, with the APU inoperative or selected OFF for performance reasons.

When below 10000 feet AGL:

(1) PRESS CONT switchAUTO

At the turn to final approach:

(2) L and R PACK switches OFF

(3) 10TH STAGE, L and R BLEED
AIRCLOSED

(4) RAM AIR switch OPEN

(5) Landing ElevationSET to 3000 feet above
landing field elevation

NOTE

1. Avoid high rates of descent for passenger comfort.
2. After landing, ensure that the airplane is completely depressurized prior to opening any airplane doors.

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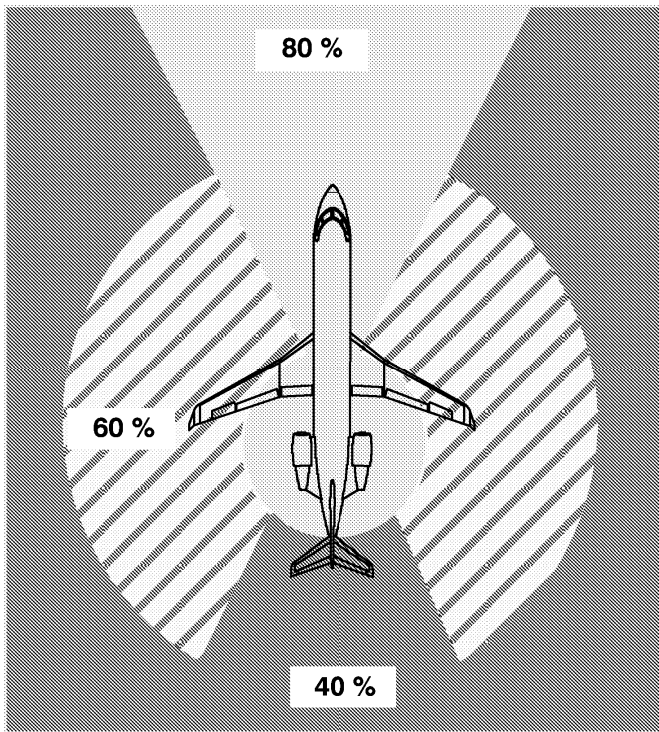
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LIMITATIONS

- Both engines’ thrust reversers must be operational.
- Fan speed must not exceed 80% N₁ reverse thrust.
- The application of brakes while the airplane is backing-up is prohibited.
- Reverse taxi operation is prohibited when ice, slush or snow is on ramp surface or during periods of heavy rain.
- Wind and fan speed limitations as detailed in the following illustration must be observed.

MAXIMUM REVERSE % N ₁	WIND CONDITION
80	Headwind / Any speed OR Any wind direction / <10 knots
60	Crosswind / >10 knots; < 40 knots
40	Crosswind / ≥ 40 knots OR Tailwind / ≥ 10 knots



NORMAL PROCEDURES

It is assumed that prior to the performance of the following procedures, the Pilot-In-Command (PIC) has ascertained that the prevailing conditions are suitable for the safe conduct of reverse taxi operations.

REVERSE TAXI PROCEDURES

After Engine Start:

- (1)

Communication

Establish
- (2)

External power / air connections

Removed
- (3)

Chocks

Removed
- (4)

Cleared-to-taxi signal

Received from ground crew
- (5)

Thrust levers

IDLE, then deploy thrust
reversers.
- (6)

Reverse thrust

INCREASE gradually and
smoothly until the airplane
moves.
- Fan speed (% N_1) must not exceed the limits established in this
supplement.

NOTE

1.

Backing-up using thrust reversers may not
always be possible. The capability to
back-up is adversely affected by increasing
airplane weight, aft centre of gravity, and/or
ramp gradient.
2.

Backing-up using thrust reversers can be
consistently accomplished on a clean, level
ramp, with airplane weights of up to
23133 kg (51000 lb), providing that
cross-winds do not limit the fan speed to
below 80% N_1 .
3.

Reverse taxi using fan speeds in excess of
60% N_1 may result in noise and vibration
levels which may not be acceptable for
passenger comfort.

- (7)

Rearward speed

Maintain by modulating the
thrust, as required.
- Reverse taxi rate must not exceed that of a normal walk.



Do not use brakes to stop the rearward movement of the airplane. If it is necessary to stop the airplane for any reason, the pilot must move the thrust levers out of reverse and stop the airplane with forward thrust.

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Reverse Taxi During Turns:

- (1) Direction of turn Confirmed with ground crew.
 - (2) Nose wheel steering tiller Set to commence turn, after signal from ground crew has been received.
- Avoid large nosewheel deflections.

Termination of Reverse Taxi:

- (1) Thrust reversers Stow and apply forward thrust.



Do not use brakes to stop the rearward movement of the airplane.

- (2) Normal taxi Commence

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DEICING / ANTI-ICING

Before spraying, if engines / APU operating:

- (1) Thrust leversIDLE
- (2) APUSee below
 - One / two step de-icing - APU as required
 - Gantry de-icing - APU shutdown
- (3) L and R PACK OFF
- (4) ANTI-ICE, WING All OFF
- (5) ANTI-ICE, LH and RH COWL All OFF
- (6) BleedsAll CLOSED
- (7) WipersOFF / PARK
- (8) Stabilizer trim Set for take-off
- (9) Flaps As required
- (10) AirplaneHead into wind, if possible
- (11) De-icing operatorAdvised
 - Engine and APU must not ingest de-icing fluid.
 - Avoid direct spray on the following:
 - Windshields, and all windows;
 - Air data probes and sensors / ports and AOA vanes;
 - Antennas;
 - Intakes and exhausts, vents and drains;
 - Winglets, wheels and brakes.
 - Type II, Type III and IV fluid MUST NEVER be applied to the following:
 - Windshields and side windows;
 - Air data probes and sensors;
 - Control surface cavities;
 - Engine and APU air intakes;
 - Lower side of radome;
 - Engine cowlings and thrust reverser components; and
 - NACA vent scoops.
 - Application of fluid should follow the sequence as follows:
 - Horizontal stabilizer,
 - Vertical stabilizer,
 - Top of fuselage
 - Sides of fuselage,
 - Wings.

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After spraying:

(12) Critical areas Inspect
to confirm complete de-icing

- Flight controls – free movement;
- Wings – upper / lower surfaces, pay particular attention to the leading edges
- Vertical and horizontal stabilizer – leading edges, upper / lower surfaces and side panels;
- Flaps, flap tracks and flap drives;
- Ailerons, elevators, rudder, spoilers and spoilerons;
- Air data probes / sensor, AOA vanes;
- Antennas;
- Fuel drains and NACA vents;
- Engine and APU intakes, APU exhaust;
- Landing gear and landing gear bays;
- Windshields, windows, door sills and surrounds.



1. Under no circumstances can an airplane that has been anti-iced, receive another coat of Type II / Type III / Type IV fluid on top of the existing film.
2. If holdover time is exceeded, surfaces must be deiced with a hot de-icing mixture of Type I fluid and water, before another application of Type II, Type III, or Type IV fluid is made.

When engines are running:

(13) Engine thrust Increase to 60% N_1 for
20 seconds to clear residual
fluid

(14) BLEED AIR, L and R 14TH
STAGE OPEN

(15) ANTI-ICE , WING As required

(16) ANTI-ICE , LH and RH COWL As required

NOTE

Wait at least 1 minute before opening the 10th stage bleed air valves and 3 minutes before turning the packs on to avoid contaminating the air-conditioning system with de-icing/anti-icing fluid. Contamination will cause objectionable fumes (causing throat irritation) and odours to enter the airplane.

- (17) BLEED AIR, L and R 10TH STAGE OPEN
- (18) L and R PACK On
- (19) Wipers As required
- (20) Take-off configuration Check

DEICING FLUID HOLDOVER TIMES

Guidelines for holdover times and de-icing fluid application are revised every year according to the Transport Canada TP 14052 and associated holdover time guideline document. The current documents are available at your country authorities website.

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INTRODUCTION

GENERAL

The Quick Reference Handbook Manual (QRH) contains limitations data for the operation of the following airplane model, manufactured by Bombardier Inc., Montreal, Quebec, Canada:

CL-600-2B19

Observance of the limitations included in this chapter is mandatory.

In the event of a disagreement between the QRH and the AFM, the AFM takes precedence.

KINDS OF AIRPLANE OPERATION

The airplane is certified in the transport category for day and night operations, in the following conditions when the equipment and instruments required by the airworthiness and operating regulations are approved, installed and in an operable condition:

- VFR and IFR
- Flight in icing conditions

The airplane is certified for ditching when the safety equipment specified by the applicable regulations is installed.

Effectivity:

- Airplanes 7004 thru 7006 and 7008 thru 7014 **not incorporating** Service Bulletin SB 601R-23-004, Communications – Audio Integrating – Installation of the Audio Electronic Unit; and
- Airplanes 7003 thru 7046 **not incorporating** Service Bulletin SB 601R-34-013, Navigation – Independent Position Determining – TCAS Aural Warning Level Adjustment.

CREW HEADSETS

All flight compartment crew members must wear headsets at all times.

The headsets approved for use are as follows:

- Sennheiser HME–1410 KA, or
- Telex Airman 750.

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STRUCTURAL WEIGHT

STRUCTURAL WEIGHT LIMITATION

Weight	kg	lb	Airplane Option Code
Maximum Ramp Weight (MRW)	23247	51250	<0002>
	24154	53250	<0004>
Maximum Take-Off Weight (MTOW)	23133	51000	<0002>
	23995	52900	<0004>
Maximum Landing Weight (MLW)	21319	47000	
Maximum Zero Fuel Weight (MZFW)	19958	44000	
Minimum flight weight	13608	30000	

NOTE

The Maximum Take-off Weight (MTOW) and/or Maximum Landing Weight (MLW) may be further limited due to performance considerations.

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OPERATING LIMITATIONS

ALTITUDE AND TEMPERATURE OPERATING LIMITS

Take-off, Landing and Operating Limits

Maximum airport pressure altitude for take-off and landing is 10000 feet.

Maximum operating altitude is 41000 feet.

Maximum ambient air temperature approved for take-off and landing is ISA + 35°C.

Minimum ambient temperature approved for take-off is -40°C (-40°F).

TAKE-OFF

Rotation Rate and Pitch Attitude

WARNING

Excessive rotation rates (exceeding 3 degrees per second) or over-rotations may lead to high pitch attitudes and angles of attack being attained while the aircraft is near the ground. This can reduce stall margins significantly, resulting in stick shaker/pusher activation and potentially loss of control. Pilots must rotate smoothly towards the target pitch attitude then transition to speed control.

Pitch Trim

WARNING

Failure to set the pitch trim appropriate to the computed centre of gravity may result in excessive rotation rate at take-off.

- Pitch trim must be set according to the airplane’s computed centre of gravity.

Effectivity:

- Airplanes **not incorporating** the -904 or the -037 Flight Control Computer:

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Take-off Pitch Target

- The initial target for rotation is 10 degrees.
- If the flight director is used for take-off, set the pitch target for 10 degrees. (Refer to Flight Crew Operating Manual, Volume 2 (CSP A-013): SUPPLEMENTARY PROCEDURES – Automatic Flight Control System – TAKE-OFF).
- Take-off performance data in the Airplane Flight Manual (CSP A-012), Chapter 6 remains applicable. (Refer to Airplane Flight Manual, (CSP A-012): – PERFORMANCE – Take-off Performance.)

OPERATION IN ICING CONDITIONS

Cowl Anti-ice System

(1) Ground Operations

The cowl anti-ice system must be ON when the OAT is 10°C (50°F) or below and visible moisture in any form is present (such as fog with visibility of 1500 metres [one mile] or less, rain, snow, sleet and ice crystals).

The cowl anti-ice system must also be ON when the OAT is 10°C (50°F) or below when operating on runways, ramps, or taxiways where surface snow, ice, standing water, or slush is present.

(2) Flight Operations

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is –40°C (–40°F) or below.

The cowl anti-ice system must be ON:

- when in icing conditions, or
- when ICE is annunciated by the ice detection system.

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Wing Anti-ice System

(1) Ground Operations

NOTE

Icing conditions exist on the ground when the OAT is 5°C (41°F) or below and:

- visible moisture in any form (such as clouds, fog or mist), is present below 400 feet AGL, or
- the runway is wet or contaminated, or
- in the presence of any precipitation (such as rain, snow, sleet or ice crystals).

Effectivity:

- Commencing 1 December 2008:
- Airplanes 7003 thru 8076, 8082, 8086, 8090 thru 8092, 8096 and 8097 **not incorporating** the following Service Bulletin:
 - SB A601R-30-032, Ice and Rain Protection – Wing Anti-Ice System – Inspection of the Wing Anti-Ice Piccolo Tubes.

Take-off in icing conditions, which would require the use of the wing anti-ice system, is prohibited.

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- The wing anti-ice system must be selected ON during final taxi prior to take-off if the OAT is 5°C (41°F) or below, unless Type II, Type III or Type IV anti-icing fluids have been applied. During single engine taxi operations, final taxi prior to take-off is defined as that period after the second engine is started.

NOTE

1. If wing anti-ice is selected ON during taxi, **L WING A/I** or **R WING A/I** caution messages may be posted but must be verified out and **WING A/ICE ON** advisory message posted prior to take-off. If wing anti-ice is not required for take-off, it should be selected OFF just prior to take-off.
2. To prevent wing contamination from reverse jet blast, operating the thrust reversers during taxi operations on wet and contaminated surfaces should be avoided.

The wing anti-ice system must be selected and confirmed ON for take-off when, the OAT is 5°C (41°F) or below and:

- visible moisture in any form (such as clouds, fog or mist), is present below 400 feet AGL, or
- the runway is wet or contaminated, or
- in the presence of any precipitation (such as rain, snow, sleet or ice crystals).

NOTE

If the wing anti-ice system is selected ON for take-off, the cowl anti-ice system must also be selected ON.

- When Type II, Type III or Type IV anti-icing fluids have been applied, the wing anti-ice system must only be selected and confirmed ON just prior to thrust increase for take-off.
- Refer to Flight Crew Operating Manual, Volume 2 (CSP A-013): SUPPLEMENTARY PROCEDURES – Cold Weather Operations. – Phase of Flight Procedures.

(2) Flight Operations

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is –40°C (–40°F) or below.

The wing anti-ice system must be ON:

- when ICE is annunciated by the ice detection system, or
- when in icing conditions and the airspeed is less than 230 KIAS.

Effectivity:

- Commencing 1 December 2008:
- Airplanes 7003 thru 8076, 8082, 8086, 8090 thru 8092, 8096 and 8097 **not incorporating** the following Service Bulletin:
 - SB A601R-30-032, Ice and Rain Protection – Wing Anti-Ice System – Inspection of the Wing Anti-Ice Piccolo Tubes.
- Continued flight in conditions requiring the use of wing anti-ice is prohibited.
- If the wing anti-ice system was selected ON in flight, leave icing conditions, and
 - If the visible portion of the wings can be confirmed, from the cockpit, to be free of ice and the **TAT** for approach and landing is greater than 10°C, accomplish a normal approach and landing.
 - If the visible portion of the wings cannot be confirmed, from the cockpit, to be free of ice and the **TAT** for approach and landing is less than or equal to 10°C, accomplish ABNORMAL PROCEDURES, Ice and Rain Protection, Wing Anti-Ice Piccolo Duct Damage Suspected Procedure.

Thrust Settings

For take-offs in icing conditions, from 0°C to 5°C OAT and at pressure altitudes of 2000 feet and below, the take-off thrust settings and associated airplane performance for 10th stage bleeds closed and engine cowl anti-icing on can be used with engine cowl and wing anti-icing bleeds switched on.

Super-cooled Large Droplet (SLD) Icing

Continued operation in areas where Super-cooled Large Droplet (SLD) icing conditions exist is prohibited.

SLD icing conditions are indicated by ice accretion on the flight compartment side windows:

- The wing anti-icing system must be ON in SLD icing conditions.
- The cowl anti-icing system must be ON in SLD icing conditions.
- Leave icing conditions when side window icing occurs.

COLD WEATHER OPERATIONS

WARNING

Even small amounts of frost, ice, snow, or slush on the wing leading edges and forward upper wing surface may adversely change the stall speeds, stall characteristics and the protection provided by the stall protection system, which may result in loss of control on take-off.

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Take-off is prohibited with frost, ice, snow or slush adhering to any critical surface (wings, horizontal stabilizer, vertical stabilizer, control surfaces and engine inlets).

Although the upper fuselage is not defined as a critical surface, it must nonetheless be de-iced to remove contamination, other than allowable frost, anytime the wing and tail surfaces require de-icing.

NOTE

1. Take-off is permitted with frost adhering to:
 - the upper surface of the fuselage; and/or
 - the underside of the wing, that is caused by cold soaked fuel,in accordance with the instructions provided in the Flight Crew Operating Manual, Volume 2 (CSP A-013): SUPPLEMENTARY PROCEDURES – Cold Weather Operations – PRE-FLIGHT PREPARATION – External Safety Inspection.
2. Comprehensive procedures for operating in cold weather are provided in the Flight Crew Operating Manual, Volume 2 (CSP A-013): SUPPLEMENTARY PROCEDURES – Cold Weather Operations.

In addition to a visual check, a tactile check of the wing leading edge, wing forward upper surface and wing rear upper surface is required during the External Walkaround inspection to determine that the wing is free from frost, ice, snow or slush when:

- the Outside Air Temperature (OAT) is 5°C (41°F) or less, or
- the wing fuel temperature is 0°C (32°F) or less, or
- the atmospheric conditions have been conducive to frost formation.

NOTE

Ice and frost may continue to adhere to wing surfaces for some time even at OAT above 5°C (41°F).

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ENHANCED TAKE-OFF PROCEDURES AND WINTER OPERATIONS TRAINING

The limitations in this paragraph are effective commencing 1 November, 2008.

No take-off shall be conducted where the OAT is 5°C (41°F) or below, unless the pilot-in-command has successfully completed specific training, within the preceding 12 calendar months, for take-off procedures, ground icing conditions and cold weather operations.

No take-off shall be conducted by a pilot where the OAT is 5°C (41°F) or below, unless that pilot has successfully completed the specific training, within the preceding 12 calendar months, for take-off procedures, ground icing conditions and cold weather operations.

If neither of the preceding requirements has been complied with, the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off, if the OAT is 5°C (41°F) or below.

Completion of the following Bombardier Aerospace course will meet the intent of this training requirement:

- Bombardier Aerospace Enhancement to Take-Off Operational Safety Margins Training.

RUNWAY SLOPES

The maximum runway slopes approved for take-off and landing are:

- +2% (uphill)
- -2% (downhill)

TAILWIND CONDITIONS

The maximum tailwind component approved for take-off and landing is 10 knots.

MINIMUM FLIGHT CREW

The minimum flight crew is one pilot and one copilot.

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CARGO

Flight must be within 40 minutes of a suitable airport, if cargo is carried in the cargo compartment. <0013> or <0059>

OPERATIONS USING QFE ALTIMETER SETTINGS

Airplane operations using QFE altimeter settings must be conducted in accordance with the Airplane Flight Manual, Chapter 7, Supplement 22, Operations Using QFE Altimeter Settings.

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POWER PLANT

ENGINES

Engine Types

Type: General Electric CF34-3B1, quantity two. <0005> or <0068>

Engine Indications

The engine limit display markings on EICAS must be used to determine compliance with the maximum/minimum limits and precautionary ranges. If EICAS markings show more conservative limits than those specified in the Engine Indications table, the limit markings on the EICAS must be used.

CF34-3B1 <0005> or <0068>			
Engine Indications			
Indication	Red (Maximum/ Minimum Limits)	Amber (Caution Range)	Green (Normal Operating Range)
N ₁ % RPM	98.6	–	0 to 98.5
N ₂ % RPM: Wing anti-ice on	99.3	0 to 77.9	78 to 99.2
N ₂ % RPM: Wing anti-ice off	99.3	–	0 to 99.2
ITT °C: Normal Take-off	900 (first 2 min.)	–	0 to 900
	884 (next 3 min.)		0 to 884
ITT °C: Go-around or APR Thrust	928 (first 2 min.)	–	0 to 928
	900 (next 3 min.)		0 to 900
ITT °C: Maximum Continuous Thrust (MCT)	900* 874 to 900† 900†	874 to 900*	0 to 874
OIL TEMP °C	163	150 to 162	–40 to 149

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CF34-3B1 <0005> or <0068>			
Engine Indications			
Indication	Red (Maximum/ Minimum Limits)	Amber (Caution Range)	Green (Normal Operating Range)
OIL PRESS psi	0 to 25	116 to 156	26 to 115
* During MCT, amber band and red tick mark appear if ITT is greater than 874°C. † During MCT, amber band becomes a red band if ITT remains above 874°C for more than 30 seconds or if ITT is greater than 900°C. Red tick mark stays at 900°C.			

Engine Warm-up

The engine must remain at IDLE until oil pressure reaches normal operating range. During all starts, do not exceed 75% N₁ for two minutes after start, or until all operating indications are in the normal range, whichever is longer.

ENGINE OPERATING LIMITS

Engine Operating Limits Table

CF 34-3B1 <0005> or <0068>			
Condition	Core RPM N ₂ %	Fan RPM N ₁ %	ITT °C
Start	20	–	900
Idle	56.5 to 68.0†	–	–
Acceleration	–	–	900
Maximum Continuous	99.2	98.6	874
Normal Take-off	98.3	96.2	884 (5 min.)* 900 (2 min. out of 5 total transient)*

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CF 34–3B1 <0005> or <0068>			
Condition	Core RPM N ₂ %	Fan RPM N ₁ %	ITT °C
APR Thrust	99.4	98.6	900 (5 min.)* 928 (2 min. out of 5 total transient)*
* Transient limits † Engine to engine N ₂ split at ground idle power should not be greater than 2% N ₂ . If N ₂ is 57% or less with an OAT of –20°C or warmer, do not accelerate above idle.			

NOTE

1. The take-off, go-around and maximum continuous thrust N₁ values for the CF34-3B1 engine are presented on the appropriate engine thrust setting charts contained in this manual. <0005> or <0068>
2. If above 40000 feet one air-conditioning unit or cowl anti-ice must be selected on for each engine.

Airplane Cold Soak

Before the first flight of a day, when the airplane is cold-soaked at an ambient temperature of –30°C (–22°F) or below for more than 8 hours:

- The engines must be motored for 60 seconds and the fan rotation must be verified before an engine start is initiated.
- Thrust reversers must be actuated until the deploy and stow cycles are less than 5 seconds.

The minimum ambient temperature for starting a cold soaked APU on the ground is –40°C (–40°F).

Oil Temperature

Min for Start	-40°C
Max Cont	+155°C
Maximum Permissible	+163°C (15 Minutes)

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Oil Pressure

Idle	25 psi minimum
Take-off	45 psi minimum
Max Cont	115 psi maximum
Max Transient (Cold Start)	156 psi maximum

CONTINUOUS ENGINE IGNITION

Continuous engine ignition must be used during the following:

- Take-offs and landings on contaminated runways;
- Take-offs with high cross-wind components;
- Flight through moderate or heavier intensity rain;
- Flight through moderate or heavier intensity turbulence;
- Flight in the vicinity of thunderstorms.

AUTOMATIC PERFORMANCE RESERVE (APR)

If take-off performance is predicated upon the use of APR, the APR system must be verified operative prior to take-off.

STARTER CRANKING LIMITS (GROUND AND AIR)

Engine Start

The starter must not be used if indicated N₂ rpm exceeds 55%.

Start	Maximum Time On	Followed By
1	1 minute	10 seconds off
2	1 minute	10 seconds off
3	1 minute	5 minutes off
4 and subs	1 minute	5 minutes off

Associated Conditions

At initiation of thrust lever movement from SHUT OFF to IDLE:

- ITT must be 120°C or less for all ground starts.
- ITT must be 90°C or less for starts. all in-flight

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Dry Motoring Cycle

The starter must not be used if indicated N₂ rpm exceeds 55%.
The dry motoring cycle may be used for:

- Engine ground starts
- Engine air starts
- Motoring of cold soaked engines

Start	Maximum Time On	Followed By
1	90 seconds	5 minutes off
2 and subs	30 seconds	5 minutes off
Thrust levers must be at SHUT OFF		

ENGINE RELIGHT

Engine starting in flight is only permitted within the envelope:

Relight Type	Envelope
Windmilling	Altitude from 21000 feet to 15000 feet: Speed 300 KIAS to V _{MO} /M _{MO} and from 12 to 55% N ₂ .
	Altitude less than 15000 feet: Speed from 300 KIAS to V _{MO} /M _{MO} and from 9 to 55% N ₂ .
Starter-assisted Cross Bleed (> 60 psi)	Altitude from 21000 feet to 15000 feet: Speed from 180 KIAS up to V _{MO} and from 0 to 55% N ₂ .
	Altitude less than 15000 feet: Speed from V _{REF} up to V _{MO} and from 0 to 55% N ₂ .

FUEL

Fuel Load

The maximum permissible fuel imbalance between the contents of the main left tank and the main right tank is 363 kg (800 lb).
Fuel remaining in a tank when the appropriate fuel quantity indicator reads zero is not usable.

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The fuel quantities that follow are based on 6.8 lb per US gal and 2.2046 lb per kg conversion factors. The maximum usable fuel load for each fuel tank is as follows:

	Pressure Refueling	Gravity Refueling
Left main tank	2159 kg (4760 lb)	2036 kg (4488 lb)
Right main tank	2159 kg (4760 lb)	2036 kg (4488 lb)
Centre tank	2267 kg (4998 lb)	2237 kg (4930 lb)
Total	6585 kg (14518 lb)	6309 kg (13906 lb)

Take-off with a fuel load in excess of 227 kg (500 lb) in the centre tank is not permitted unless:

- each main wing tank is above 1996 kg (4400 lb); or
- each main wing tank is above 907 kg (2000 lb): and
 - the allowable zero fuel weight is reduced by the weight of the fuel in the centre tank in excess of 227 kg (500 lb); and
 - the Centre of Gravity (CG) in this configuration is verified to be within the allowable CG envelope as calculated from the Weight and Balance Manual (WBM) (CSP A-041).

The minimum fuel quantity for go-around is 202 kg (450 lb) per wing (with the airplane level) and assuming a maximum airplane climb attitude of 10 degrees nose up.

Fuel Temperature

Take-off with engine fuel temperature indications below 5°C (41°F) is prohibited.

Take-off with bulk fuel temperature indications below the limits stated is prohibited.

During flight, bulk fuel temperature must remain above the applicable bulk fuel freezing point.

Fuel Type	Bulk Fuel Take-off Limit – Minimum Temperature	Bulk Fuel Freezing Point
ASTM D1655 JET A	–30°C	–40°C
ASTM D1655 JET A-1	–37°C	–47°C
MIL-DTL-5624 JP-5	–36°C	–46°C
MIL-DTL-83133 JP-8	–37°C	–47°C

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Fuel Type	Bulk Fuel Take-off Limit – Minimum Temperature	Bulk Fuel Freezing Point
GB 6537-2006 No. 3 Jet	–37°C	–47°C
CIS TS-1	–43°C	See NOTE
CIS RT	–40°C	
Ukrainian TS-1	–42°C	–52°C
Ukrainian RT	–42°C	–52°C

NOTE

- 1. Russian / CIS TS-1 and RT fuels with a freezing point of not higher than –50°C are approved for use wherein the ground level OAT is not below –30°C during the 24 hours before departure.
- 2. Russian / CIS fuel with a freezing point of not higher than –55°C (RT) and –60°C (TS-1) for use in low temperature regions are available at operator’s request.
- 3. The principle grade available in Russia (and members of the CIS) is TS-1 (written as TC-1 in Russian cyrillic script).

Use of wide-cut fuel is prohibited except for non-revenue ferry flights, with the following fuel temperature limitations:

Fuel Type	Bulk Fuel Take-off Limit Minimum Temperature	Bulk Fuel Maximum Operating Temperature	Bulk Fuel Freezing Point
ASTM D6615 Jet B	–40°C	+45°C	–50°C
MIL-DTL-5624 JP-4	–48°C	+45°C	–58°C

Powered cross-flow is prohibited when using wide-cut fuel and the APU is to be operated.

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Fuel Grades

Fuels conforming to any of the following specifications are approved for use. Mixing of fuels is permitted.

Canadian	American	British	Chinese	CIS/ Ukrainian	NATO
CGSB-3. 23	ASTM D1655 JET A				
	ASTM D1655 JET A-1	DEF STAN 91-91	GB 6537- 2006 No. 3 Jet	RT TS-1☆	F-35
CGSB-3. 24	MIL-DTL- 83133 JP-8	DEF STAN 91-87			F-34
	MIL-DTL- 5624 JP-5	DEF STAN 91-86			F-44
CIS/Ukrainian Fuels Inspection Requirement ☆ Engine fuel system components must be inspected in compliance with SB 601R-73-009, CF34-3A1 and CF34-3B1 — Russian Fuel Fleet Leader Sampling Program.					

Fuel Additives

- Alcohol-based anti-icing additives of Fuel System Icing Inhibitors (FSII) type Fluid I and Fluid I-M should not be used continuously. Russian / CIS fuel containing (or pre-mixed with) alcohol-based additives of anti-icing Fuel System Icing Inhibitor (FSII) type Fluid I and Fluid I-M should not be used continuously. Contact engine manufacturer for details of impact on a reduced component inspection program if continuous use has been determined.
- Anti-icing additives Diethylene Glycol Monomethyl Ether (DiEGME) per MIL-DTL-85470:

GEAE Fuel Specification D50TF2 Class and Fuel Description	Additive Maximum Concentration
Class A: Aviation Kerosene (Jet A)	0.15%
Class B: Wide-cut Distillate (Jet B, JP-4)	0.15%
Class C: Low Freeze Kerosene (Jet A-1, JP-8, No. 3 Jet)	0.15%

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GEAE Fuel Specification D50TF2 Class and Fuel Description	Additive Maximum Concentration
Class D: High Flash Kerosene (JP-5)	0.15%
Class E: Low Freeze Kerosene (CIS RT)	0.20%

- Kathon FP 1.5 biocide additive at a concentration of not more than 100 parts per million (ppm) by weight for the initial dose, to prevent the growth of micro-organisms. Contact the additive manufacturer for specific use instructions. The maintenance dose should be as per the additive manufacturer’s instructions and should not exceed 50 ppm.
- SOHIO Biobor JF biocide additive at a concentration of not more than 270 ppm (20 ppm elemental boron) by weight for the initial dose, to prevent the growth of micro-organisms. Contact the additive manufacturer for specific use instructions. The maintenance dose should be as per the additive manufacturer’s instructions and should not exceed 135 ppm.
- Anti-static additive STADIS 450 at a concentration of 3 g/m³ for the initial dose. Maximum dosage should not exceed 5 g/m³.
- Fuel system additive SIGBOL may be used in concentrations less than 0.0005% by weight. This additive should not be used continuously. Contact engine manufacturer for details of impact on the component inspection program.
- Corrosion inhibitors listed in the table that follows are approved by the concentrations indicated. It is recommended that corrosion inhibitors be blended with the fuel to provide lubricity. The corrosion inhibitor must be added after water removal and downstream of any clay filters (these processes remove the inhibitor).

Additive	Maximum Concentration
Apollo PRI-19	23 g/m ³
Octel DCI-4A	23 g/m ³
Octel DCI-6A	6 g/m ³
Hitec E-580	23 g/m ³
Nalco 5403	23 g/m ³
Nalco 5405	11 g/m ³

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Fuel Cross-flow

Powered cross-flow and gravity cross-flow must be off for take-off.

OIL

Oil Grades

NOTE

Mixing of different types of oils is prohibited.

Component	MIL-L-23699 (Type II)	MIL-L-7808 (Type I)	Castrol 4000
Engines	√	√	√
Air Turbine Starter	√	√	√
Integrated Drive Generator	√	√	√
Auxiliary Power Unit	√	√	√
Air Cycle Machines	√	√	√

Oil Consumption

Maximum oil consumption, on each engine, is 189 cubic centimetres per hour (6.4 US ounces per hour / 0.05 US gallon per hour).

Engine Oil Level

Engine oil levels must be checked as follows:

- The maximum duration without engine oil servicing is 16 operating hours.

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- For airplane operations in excess of 16 operating hours (without engine oil tank servicing), the engine oil level must be checked from the flight compartment ENGINE OIL LEVEL panel within 3 minutes to 2 hours after every engine shutdown.

NOTE

The engine oil level panel is used with the engines not running. Erroneous indications will occur if utilized during any other condition, as stated above.

Oil Replenishment System

If use of the oil replenishment system is required, then the engine(s) should be replenished within 15 minutes to 2 hours after engine shutdown.

Maximum refill allowable is 1890 cubic centilitres (2 US quarts) without dry motoring the engine.

If the oil system has to be replenished to maximum capacity and the replenishment period has been exceeded, the engine(s) must be dry motored.

AUXILIARY POWER UNIT (APU)

Type

Garret GTCP-36-150RJ

Maximum RPM

107%

Maximum EGT

743°C

Starting

Minimum ambient temperature for starting a cold soaked APU on the ground is -40°C.

The following APU start cycles are permitted:

- Using airplane batteries on the ground or for normal in-flight start:

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Start	Maximum Time On	Followed By
1	30 seconds	–
2	30 seconds	20 minutes off
3	30 seconds	–
4	30 seconds	40 minutes off

- Using ground power:

Start	Maximum Time On	Followed By
1	15 seconds	–
2	15 seconds	20 minutes off
3	15 seconds	–
4	15 seconds	40 minutes off

- If, in either case a successful start is not obtained, a further start must not be attempted for a period of at least 40 minutes.

NOTE

It is recommended that a 2-minute delay be observed between cranking attempts to allow for cooling of starter and start contactor for APU drainage.

Maximum EGT:

- 974°C (not to be exceeded under any operating conditions).

APU Bleed Air

APU Bleed Air Limitations	
System/Condition	Limitation
Air-conditioning	Bleed air extraction from the APU is not permitted above 15000 feet.

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APU Bleed Air Limitations	
System/Condition	Limitation
Engine-start during ground operations	No bleed air extraction limitation. Each engine may be started using the APU as a bleed air source. If both engines are to be started using the APU bleed air, then the operating engine's thrust must not exceed 70% N ₂ .
Engine-start during flight	During single engine operations, APU bleed air extraction for an engine start is not permitted. During double engine failure conditions, APU bleed air extraction for engine starts is permitted. However, APU bleed air extraction is limited to 13000 feet and below.

APU Generator

The maximum permissible load on the APU generator in flight is 30 kVA.

APU Indications

The APU limit display markings on the EICAS must be used to determine compliance with the maximum/minimum limits and precautionary ranges. If EICAS markings show more conservative limits than those specified in the APU Indications table, the limit markings on the EICAS should be used.

APU Indications			
Indication	Red (Maximum/ Minimum Limits)	Amber (Caution Range)	Green (Normal Operating Range)
APU EGT °C	Greater than 743	713 to 742	0 to 712
APU RPM %	Greater than 107	101 to 106	0 to 100

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OPERATING SPEEDS

MAXIMUM OPERATING SPEED AND MACH NUMBER

Maximum operating limit speeds, must not be deliberately exceeded in any regime of flight (climb, cruise or descent), unless a higher speed is specifically authorized for flight test or training operations.

DESIGN MANEUVERING SPEED

Full application of rudder and aileron controls as well as maneuvers that involve angles of attack near the stall, must be confined to speeds below V_A .



Avoid rapid and large alternating control inputs, especially in combination with large changes in pitch, roll, or yaw (e.g., large side slip angles) as they may cause structural failure at any speed, including below V_A .

Effectivity:

- Airplanes **incorporating** the following Service Bulletin:
 - SB 601R-11-090, Placards and Markings – Flight Compartment – Installation of the New Maximum Flaps Operating Speed Placard.

MAXIMUM FLAPS OPERATING SPEED

The maximum speeds at which it is safe to extend the flaps are:

- FLAPS 0 to 8: 200 KIAS
- FLAPS 8 to 20: 200 KIAS
- FLAPS 20 to 30: up to V_{FE} for FLAPS 30
- FLAPS 30 to 45: up to V_{FE} for FLAPS 45

The maximum speed at which it is safe to retract the flaps is V_{FE} for the initial flaps position.

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MAXIMUM FLAPS EXTENDED SPEED

The maximum speeds at which the airplane may be flown with the flaps extended are:

- FLAPS 8: 230 KIAS
- FLAPS 20: 230 KIAS
- FLAPS 30: 185 KIAS
- FLAPS 45: 170 KIAS

Effectivity:

- Airplanes 7003 thru 7319 **not incorporating** the following Service Bulletin:
- SB 601R-11-071, Placards and Markings, Reduced V_{FE} , or
- SB 601R-11-076, Placards and Markings, Reduced V_{FE} , or
- SB 601R-11-080, Placards and Markings – Flight Compartment – Installation of the New Airspeed Limitation Placard and Removal of the Flap Inspection Placard.

The maximum speeds at which the airplane may be flown with the flaps extended are:

- FLAPS 8: 230 KIAS
- FLAPS 20: 230 KIAS
- FLAPS 30: 196 KIAS
- FLAPS 45: 191 KIAS

Effectivity:

- Airplanes 7003 thru 7319 **incorporating** the following Service Bulletin:
- SB 601R-11-071, Placards and Markings, Reduced V_{FE} , or
- SB 601R-11-076, Placards and Markings, Reduced V_{FE} , or

The maximum speeds at which the airplane may be flown with the flaps extended are:

- FLAPS 8: 215 KIAS
- FLAPS 20: 215 KIAS
- FLAPS 30: 185 KIAS
- FLAPS 45: 170 KIAS

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Effectivity:

- Airplanes 7320 thru 7903 **not incorporating** the following Service Bulletin:
 - SB 601R-11-080, Placards and Markings – Flight Compartment – Installation of the New Airspeed Limitation Placard and Removal of the Flap Inspection Placard.

The maximum speeds at which the airplane may be flown with the flaps extended are:

- FLAPS 8: 215 KIAS
- FLAPS 20: 215 KIAS
- FLAPS 30: 185 KIAS
- FLAPS 45: 170 KIAS

MAXIMUM LANDING GEAR OPERATING SPEED

The maximum airspeed at which it is safe to extend the landing gear is 250 KIAS.

The maximum airspeed at which it is safe to retract the landing gear is 200 KIAS.

MAXIMUM LANDING GEAR EXTENDED SPEED

The maximum airspeed at which the airplane may be flown with the landing gear extended and locked is 250 KIAS.

TIRE LIMIT SPEED

The tire limit speed is 182 knots ground speed.

MAXIMUM AIRSPEED FOR ADG OPERATION

Effectivity:

- Airplanes 7002 thru 7304:

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The maximum airspeeds for continuous operation of the air-driven generator are as follows:

- Following automatic or manual deployment, speeds up to 250 KIAS are permitted for the remainder of the flight.
- When operationally necessary (e.g. windmill airstarts) speeds up to 330 KIAS are permitted for a period of twelve (12) minutes, and speeds between 331 and 335 KIAS are permitted for a period of four (4) minutes.
- For flight test or pilot training deployment, the maximum airspeed is 215 KIAS.

TURBULENCE PENETRATION SPEED

Maximum airspeed for turbulence penetration is 280 KIAS or 0.75 Mach, whichever is lower.

MINIMUM OPERATING LIMIT SPEED

Intentional speed reduction below the onset of stall warning, as defined by stick shaker operation, is prohibited unless a lower speed is specifically authorized for flight test or training operations.

WINDSHIELD WIPER (IF INSTALLED)

The windshield wiper must not be operated above 220 KIAS.
If the windshield wiper has failed in a non-parked position, the airplane speed must not exceed 220 KIAS.

MANEUVERING LOADS

MANEUVERING LIMIT LOAD FACTORS

These load factors limit the permissible angles of bank in turns and the severity of pull-up and push-over maneuvers:

- Flaps up: –1.0 g to 2.5 g
- Flaps down: 0.0 g to 2.0 g

SIDE-SLIP MANEUVERS

Avoid unnecessary and large side-slip maneuvers during low-speed, high-altitude cruise.

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SYSTEM LIMITATIONS

AIR-CONDITIONING AND PRESSURIZATION

The maximum positive differential pressure is 8.7 psi.

The maximum negative differential pressure is –0.5 psi.

During ground maneuvering, the pressure differential must not exceed 0.1 psi.

During initial landing (at touchdown), the maximum differential pressure must not exceed 1.0 psi.

The airplane must be completely depressurized prior to opening any of the airplane doors.

The cabin pressurization system must not be operated to –1500 feet when the system is in manual mode.

AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)

Operation with the autopilot engaged is prohibited at altitudes below 600 feet AGL, except when performing the following:

- For visual and non-precision approaches, the minimum autopilot use height is 400 feet AGL.

Effectivity:

- Airplanes **incorporating** the –404 or the –604 Flight Control Computer:
- For precision approaches, the minimum autopilot use height is 60 feet AGL. <JAA>
- Use of IAS mode (flight director or autopilot coupled) during approach is not permitted.
- APPR mode VOR approaches (flight director or autopilot coupled) are not permitted.

Effectivity:

- Airplanes **incorporating** the –704 or the –037 Flight Control Computer:
- For precision approaches, the minimum autopilot use height is 80 feet AGL. <JAA>
- Use of DES submode (flight director or autopilot coupled) during approach is not permitted.

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BLEED AIR SYSTEMS

The bleed air 10th stage valves must be closed for take-off and landing if the engine cowl and/or wing anti-ice systems have been selected on.

ELECTRICAL SYSTEMS

Permissible Loads on AC System

Individual AC generator loading must not exceed the following values:

Altitude (Feet)	Load Limitation (kVA)	
	Main Generator (Each)	APU Generator
0 – 35000	30	30
35001 to 37000	25	30
37001 and above	25	0

Permissible Loads on DC Systems

(1) Ground Operations

To protect the flight compartment CRT displays, the maximum permissible time for ground operations with DC power only is 5 minutes.

(2) Flight Operations

The maximum permissible continuous load on each TRU is 100 amp.

Circuit Breaker Reset (In Flight)

A circuit breaker must not be reset or cycled (i.e., opened or closed) unless doing so is consistent with explicit procedures specified in this AFM or unless, in the judgement of the Pilot-In-Command (PIC), that resetting or cycling of the circuit breaker is necessary for the safe completion of the flight.

FLIGHT CONTROLS – LIFT/DRAG DEVICES

Flaps

En-route use of flaps is prohibited.

Flight with flaps extended at altitudes above 15000 feet is prohibited.

Effectivity:

- Airplanes 7904 and subsequent; and
- Airplanes **incorporating** the following Service Bulletin:
 - SB 601R-11-080, Placards and Markings – Flight Compartment – Installation of the New Airspeed Limitation Placard and Removal of the Flap Inspection Placard.
- Except for touch-and-go training, take-off flaps must be selected from the 0 detent.
- Touch-and-go training is not permitted in cross-winds greater than 14 knots.

Flight Spoilers

Flight spoilers must not be extended in flight below an altitude of 300 feet AGL.

To ensure adequate maneuver margins, flight spoilers must not be extended in flight at airspeeds below the recommended approach speed plus 17 KIAS (refer to Chapter 6, PERFORMANCE – Landing Performance).

Effectivity:

- Airplanes 7003 thru 7112 **not incorporating** the following Service Bulletin:
 - SB 601R-27-053, Replacement of the Horizontal Stabilizer Trim Control Unit.

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Mach Trim

Use of Mach trim is prohibited.

If the autopilot is disengaged or inoperative, speed must be restricted to 250 KIAS (0.7 M).

NOTE

Pending incorporation of Alert Service Bulletin A601R-27-054, Flight Controls – Horizontal Stabilizer Control System - Deactivation of the Mach Trim Engage / Disengage Switch/light, the MACH TRIM caution message will be displayed on EICAS and the Mach Trim Engage / Disengage Switch "INOP" light will be illuminated (on). The EICAS message can be scrolled out of view prior to take-off, but the switch "INOP" light will remain illuminated (on).

Effectivity:

- Airplanes that **do not comply** with Transport Canada Airworthiness Directive CF-2002-35R2, Aileron Stiffness, Part C – Termination of Control Checks:

Ailerons

When departing from a wet or contaminated runway:

- at 10000 feet MSL, or when the **SAT** is less than or equal to 0°C, whichever occurs later;
- the ailerons must be operated manually every 5000 feet until the top of climb.

STALL PROTECTION SYSTEM

- The Stall Protection System test must be accomplished before the first flight of the day. (Refer to Flight Crew Operating Manual, Volume 2 (CSP A-013): NORMAL PROCEDURES – Prior to Start – ORIGINATING CHECK – STALL Test as per FCOM TR RJ/179 or later applicable revision.)
- Both stall protection system switches must remain on for all phases of flight.

THRUST REVERSERS

Thrust reversers are approved for ground use only.

The thrust reversers are intended for use during full stop landings. Do not attempt a go-around maneuver after deployment of the thrust reversers.

Take-off with any of the following thrust reverser lights on, icons or EICAS messages displayed is prohibited:

- THRUST REVERSER UNLK light,
- REV icon on N₁ gauge,
- **L REV UNLOCKED** or **R REV UNLOCKED** caution message.

During landing, application of the reverse thrust is permitted as follows:

- From touchdown speed to 60 KIAS - maximum N₁,
- Below 60 KIAS, reverse thrust must be reduced to idle.

During pre-flight check of the thrust reversers with the airplane stationary, reverse thrust must be limited to reverse idle.

NOSEWHEEL STEERING SYSTEM

Towbarless towing is prohibited, unless the operation is performed in compliance with the Aircraft Maintenance Manual (AMM) (CSP A-001) towbarless towing requirements.

TAXI LIGHTS

The taxi lights must be switched OFF whenever the airplane is stationary in excess of 10 minutes.

WHEEL BRAKE COOLING LIMITATIONS

- Brake cooling times (established in accordance with PERFORMANCE – Take-off Performance – MAXIMUM ALLOWABLE BRAKE TEMPERATURE FOR TAKE-OFF) must be observed between a landing or a Rejected Take-Off (RTO) and a subsequent take-off, to ensure that sufficient brake energy is available to bring the airplane to a complete stop, if the subsequent take-off is rejected.

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TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS)

The use of **TCAS** must comply with the appropriate national operational regulations. <JAA>

Effectivity:

Airplanes 7891 and subsequent, or

Airplanes 7003 thru 7890 **incorporating** the following Service Bulletin:

SB 601R-34-139, Navigation – Integrated Standby Instruments System – Replace the Standby Attitude Indicator (SAI) and the Standby Altimeter/Indicated Airspeed (SA/IAS) Indicator with the Integrated Standby Instrument (ISI), or

SB 601R-34-137, Navigation – Integrated Standby Instrument System (ISIS) – Installation of the Thales Standby Instrument (ISI).

INTEGRATED STANDBY INSTRUMENT (ISI)

Flight Operations

When NAV 1 is tuned to a valid ILS frequency, the ISI will display localizer and glide slope deviation while on the backcourse approach. Use of the ISI localizer and backcourse information is prohibited during backcourse approaches.

CONFIGURATION DEVIATION LIST (CDL)

If the airplane is to be operated with certain secondary airframe and/or any nacelle parts missing, operation must be in accordance with the limitations specified in the basic Airplane Flight Manual, and as amended by the Configuration Deviation List (Appendix 1).

ENHANCED GROUND PROXIMITY WARNING SYSTEM (**EGPWS**)

<0040>

Airplane navigation must not be predicated upon the use of the terrain display.

To avoid giving unwanted alerts, the terrain awareness alerting and display functions must be inhibited on take-off, approach or landing within 15 nm of an airport not included in the EGPWS airport database.

The terrain data base, terrain displays and alerting system do not account for man-made obstructions, except for all known man-made obstructions in Canada, the United States and Mexico.



Effectivity:

- Airplanes **equipped** with GPS:<0027> or <0047>

1. If GPS is inoperative (identified by NO GPS RAIM, GPS NOT AVAILABLE or GPS – FMS DISAGREE message on the FMS CDU), the EGPWS may fail to alert when a terrain threat exists, during ambient surface temperatures below ISA, during non-standard temperature lapse rates or if incorrect barometric settings are set on the altimeter. <0040>

Effectivity:

- Airplanes **incorporating** the following Service Bulletin:
- SB A601R-29-031, Hydraulic Power – Hydraulic System No. 3 – Deactivation and Removal of the Accumulator Assembly, and
- Airplanes **not incorporating** the following Service Bulletin:
- SB 601R-29-034, Hydraulic Power – Hydraulic System No. 3 – Installation in a New Location of the New Accumulator and the Nitrogen Pressure Gauge of Hydraulic System No. 3.

HYDRAULIC 3B PUMP

The Hydraulic 3B pump switch must be selected ON instead of AUTO for all phases of flight.

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AIR TRAFFIC CONTROL TRANSPONDER, ENHANCED MODE S
<0096>

The installed Mode S system satisfies the data requirements of ICAO Doc. 7030/4 “Regional Supplementary Procedures for SSR Mode S Enhanced Surveillance in Designated European Airspace”. The following table represents system capability to transmit data parameters:

ATC Transponder, Enhanced Mode S Transmission Capability	
Parameter	Available / Not Available
Magnetic Heading <0025>	Available (See NOTE)
Indicated Airspeed	Available
Mach Number	Available
Vertical Rate	Inertial rate of climb/descent is available. <0025>
Roll Angle	Available
Track Angle Rate <0025>	Available
True Track Angle <0024> or <0049> or <0050>	Available
Ground Speed <0024> or <0049> or <0050>	Available
Selected Altitude	Available
Barometric Pressure Setting	Available

NOTE

MAG heading must be selected for display on the Primary Flight Display. If TRUE heading is selected, the Enhanced Mode S Transponder will downlink true heading. <0025>

NAVIGATION SYSTEM LIMITATIONS

FLIGHT MANAGEMENT SYSTEM (FMS) <0024> or <0049> or <0050>

Operating Data <0024> or <0049> or <0050>

The Flight Management System (FMS) must be operated in accordance with the latest edition of the following:

- Airplane Flight Manual, and
- Flight Management System Pilot Guide that follows:

Publication Number	Applicability	
523-0778363	FMS-4200	<FMC-002, FMC-006, FMC-011, and FMC-013>

FMS-4200 Operating Limitations <0024> or <0049> or <0050>

(1) FMC-002, FMC-006, FMC-011, and FMC-013

- Software Program Version
 - The FMS is approved for use only with the software program version given in the table that follows:

Software Program Version (SCID)	Applicability
829-7739-008	<FMC-002>
829-7739-012	<FMC-006>
832-4119-012	<FMC-011>
829-7739-028	<FMC-013>

- AC 20-130A
 - The FMS meets the enroute, terminal and non-precision approach lateral performance / accuracy criteria of AC 20-130A.

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LIMITS 8-2

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- RNP-10 Operations <0024> and <0025>
 - The dual FMS installation, when used in conjunction with the dual Inertial Reference System (IRS), meets the requirements of FAA Order 8400.12A, Rev. A, dated January 2, 1998 - Required Navigation Performance (RNP-10) Operational Approval, as a primary means of navigation for flights up to 6.2 hours in duration, without updating. The determination of flight duration starts when the system is placed in the navigation mode.

NOTE

The demonstration of performance with the above criteria does not constitute approval to conduct RNP-10 operations.

- AC 90-100A US Terminal and Enroute Area Navigation (RNAV) Operations
 - When equipped with an operating FMS and GPS, the aircraft meets the functional and the ± 1 nm accuracy requirements of AC 90-100A US Terminal and Enroute Area Navigation (RNAV) Operations.
 - AC 90 – 100A operations must not be conducted unless all of the required equipment specified below is operational

AC 90–100A Required Equipment List

Equipment	Requirements for AC 90–100A
Flight Management Computer (2) <0024>	One (1) must be operational.
FMS Control Display Unit (2) <0024>	One (1) must be operational.
Global Positioning System (2) <0027>	One (1) must be operational.

NOTE

Compliance with the standard noted above does not constitute an operational approval.

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- TGL-10 PRNAV Operations
 - The FMS installation meets the airworthiness certification requirements of JAA Temporary Guidance Leaflet No. 10, Airworthiness and Operational Approval for Precision RNAV Operations in Designated European Airspace.
 - Precision RNAV operations must not be conducted unless:
 - all of the required equipment specified in the PRNAV Required Equipment List table is operational,

PRNAV Required Equipment List	
Equipment	Requirements for PRNAV
Flight Management Computer	One (1) must be operational.
FMS Control Display Unit	One (1) must be operational and controlling an operational FMC.
VHF NAV, DME, GPS	One (1) VHF NAV and one (1) DME must be operational, or one (1) GPS is operational.

- the following messages are not displayed on any PFD or CDU:
 - FMS DR
 - IRS ONLY <0025>
 - VOR/DME DIST >75 NM
 - FMS NAV INVALID
- ILS, LOC, LOC-BC, LDA, SDF and MLS approaches using the FMS are prohibited. The FMS does not provide automatic transition to ILS.
- The FMS is approved for use only during enroute, terminal and non-precision approach phases of flight.
- FMS thrust, range, fuel management and altitude/speed capability information is advisory only.
- IFR enroute and terminal navigation is prohibited unless the pilot verifies the currency of the database and verifies waypoints for accuracy by reference to current publications.
- Instrument approaches must be accomplished with instrument approach procedures that are retrieved from the FMS database. The FMS must incorporate the current update cycle. The pilot must verify approach waypoints for accuracy by reference to current publications.



LIMITS 8-4

Rev. 61, Oct 01/2014

- The area navigation system used for IFR Class 1 navigation meets the performance/accuracy criteria of AC 20-130A for enroute and terminal area navigation.
- The approach must not be continued past the final approach fix if an FMS “NO APPR” message is displayed on the PFD.
- The VNAV system information is not temperature compensated. FMS generated altitudes and V-paths are not corrected for non-ISA conditions.
- The performance database must be verified to be current and valid for the appropriate engine model installed on the airplane:

Engine Model	Performance Database Part Number	Applicability
CF34-3B1 (Basic) <0005> and <0039>	832-4172-103	<0039>

- The FMS calculated thrust setting must not be used if the pressure altitude is greater than 36000 feet. <0039>
- Editing of the runway extension waypoint distance during a synchronized visual approach is prohibited.
(FMC-002 only) <0024>

GLOBAL POSITIONING SYSTEM (GPS) <0027> or <0047>

General

Other approved navigation equipment appropriate to the route of flight (enroute and terminal) must be installed and operating.

The GPS may only be used for approach guidance if the reference coordinate data system for the instrument approach is WGS-84 or NAD-83.

RNP-10 Operations <0024> and <0027>

The dual GPS installation, when used in conjunction with the Prediction Program 832-3443-005 or later applicable revision, and with two operational FMS-4200 systems, has been found to comply with the requirements of MNPS operations and as a primary means of navigation in oceanic and remote airspace operations. Likewise, this compliance is considered as meeting RNP-10 requirements, without time limitations.

NOTE

The demonstration of performance with these criteria does not constitute approval to conduct MNPS, oceanic/remote airspace or RNP-10 operations.

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HEAD-UP GUIDANCE SYSTEM <0026>

HEAD-UP GUIDANCE SYSTEM LIMITATIONS

Effectivity:

- Airplanes **equipped** with the (-503) HGS Computer:

NOTE

Data provided herein for AFM chapters 1 through 6 cover use of HGS in operations not less than Category I minima.

The following table identifies the combinations of mode and flight phase of the HGS which have been certified.

- An "X" indicates the combination is prohibited from use;
- A "✓" indicates no restriction unless accompanied by a limiting footnote;
- A footnote indicates an additional reference or a conditional authorization.

Limitations on the Use of HGS <JAA>

	Flight Phases					
Selected HGS Mode	Take-off	Climb	Cruise	Descent	Approach	Landing and Roll-out
PRI	✓ [1], [2]	✓	✓	✓	✓ [6]	✓ [12], [13]
VMC [7]	X	X	✓	✓	✓ [3]	✓ [12], [13]
FD	X	X	X	X	X	X
AI	X	X	X	X	✓ [3], [4], [9]	✓ [12], [13]
AII	X	X	X	X	✓ [3], [4], [9], [10]	✓ [12], [13]

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LIMITS 9-2

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Limitations on the Use of HGS <JAA>

Selected HGS Mode	Flight Phases						Landing and Roll-out
	Take-off	Climb	Cruise	Descent	Approach		
AIII	X	X	X	X	✓ [3], [5], [8], [9], [11]		✓ [8],[12]

General

- The system must be operated in accordance with the HGS Pilot Guide P/N 9701-0971 Dated May 1999 or later applicable edition.

Excessive or Unusual Attitudes

- In the event of loss of situation awareness or an upset leading to the occurrence of an unusual attitude, attempted recovery with HGS is prohibited. Refer to the PFDs.

Low Visibility Take-off

- The HGS take-off mode has been shown to meet the requirements of FAR-AWO Subpart 4. Airplanes equipped with HGS may be permitted by the operators' local authority to take-off in an RVR of 75 metres or more.

- [1] Rolling take-offs requiring localizer information are prohibited.
- [2] Take-offs using back course localizer information are prohibited.
- [3] Non-precision approaches are prohibited.
- [4] Glide slope angles outside the range of – 2.5 to –3.5 degrees are prohibited.
- [5] Glide slope angles outside the range of – 2.5 to –3.0 degrees are prohibited.
- [6] Except for go-around, use of PRI mode below 800 feet AGL is prohibited.
- [7] Use of VMC mode in IMC is prohibited.
- [8] Use of flaps other than 45 is prohibited.
- [9] Use of any A mode to monitor autopilot-coupled approaches is prohibited.
- [10] Additional limitations and procedures for AII mode are covered in Supplement 4.
- [11] Additional limitations and procedures for AIII mode are covered in Supplement 8.
- [12] Landing with the combiner deployed is prohibited if the standard airplane sun visor is not stowed.
- [13] The landing and roll-out must be done visually, however, it is permissible for the HGS combiner to be left deployed and for the mode symbology to be displayed.

REQUIRED EQUIPMENT LIST

REQUIRED EQUIPMENT LIST

Category II Required Equipment List

An ILS approach to Category II minima must not be commenced or continued unless all required airborne equipment, specified in the following Category II Required Equipment List and their ground installations, are operating satisfactorily.

Category II Required Equipment List	
Equipment	Prior to Approach
VHF NAV 1 and 2	Both must be operational.
VHF COM 1 and 2	Both must be operational.
PFD 1 and 2	One (1) PFD available and operational for each side.
Stab Channel 1 and 2	One (1) channel must be operational.
Radio Altimeter's	Both must be operational, with display on both sides.
IRS 1 and 2	Both must be operational.
ADC 1 and 2	Both must be operational.
FD 1 and 2	Both must be operational.
EFIS Comparator Monitors	Must be operational.
AFCS PITCH TRIM	Must be operational.
Autopilot	Must be operational.
Hydraulics (3)	All systems must be operational.
Electrics	Two (2) generators on and sharing load, i.e., two (2) main generators or one (1) main generator and APU generator on.

Category IIIA Required Equipment List <0026>

The following airplane equipment is required to be operative.

Category IIIA Required Equipment List	
Equipment	Quantity
Head-up Guidance System	One
VHF Navigation Receiver	Two

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Category IIIA Required Equipment List	
Equipment	Quantity
VHF Communication Receiver	Two
Primary Flight Display	Two
Stabilizer Channel	One
Radio Altimeter	Two
Inertial Reference System	Two
Air Data Computer	Two
Generator	Two
Flight Directors	Two
Yaw Dampers	One
Ground proximity Warning System	One

BRNAV Required Equipment List

The following equipment is required to be operative.

BRNAV Required Equipment List	
Equipment	Quantity
VOR	One
DME	One
FMC	One, same side
FMC CDU	One, same side
FMS Database	Current
MFD FMS MAP Mode	One (on PF side)
EFIS Control Panel	One (on PF side)

BRNAV shall not be continued if an FMS “CHECK POSITION” message is displayed.

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TAKE-OFF

Stabilizer Trim Setting

Stabilizer Trim Setting for FLAPS 8 or 20 Take-off

C.G. %MAC	All Weights
9	8.2
10	8.0
12	7.7
14	7.4
16	7.1
18	6.7
20	6.4
22	6.1
24	5.8
26	5.4
28	5.1
30	4.8
32	4.5
34	4.2
35	4.0

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Normal Take-off Thrust Setting

Normal Take-off Thrust Setting, %N₁ Zero Engine Bleeds <0005>

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING ZERO ENGINE BLEED CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	80.9	81.3	82.1	82.9	83.9	84.9	85.9
-35	-31	81.8	82.1	83.0	83.8	84.8	85.8	86.8
-30	-22	82.6	83.0	83.8	84.7	85.6	86.6	87.7
-25	-13	83.4	83.8	84.7	85.5	86.5	87.5	88.6
-20	-4	84.2	84.7	85.5	86.4	87.4	88.4	89.5
-15	5	85.1	85.5	86.3	87.2	88.2	89.3	90.3
-10	14	85.9	86.3	87.2	88.0	89.1	90.1	91.2
-5	23	86.7	87.1	88.0	88.9	89.9	91.0	92.0
0	32	87.5	87.9	88.8	89.7	90.7	91.8	92.9
5	41	88.3	88.7	89.6	90.5	91.5	92.6	93.3
10	50	89.0	89.5	90.4	91.3	92.4	92.7	92.7
15	59	89.8	90.3	91.1	92.1	92.2	92.2	92.1
20	68	90.6	91.0	91.7	91.7	91.7	91.7	91.6
25	77	91.3	91.3	91.3	91.3	91.3	91.2	91.0
30	86	90.9	90.9	90.9	90.8	90.7	90.5	90.3
35	95	90.2	90.3	90.3	90.1	90.0	89.8	89.6
40	104	89.4	89.4	89.5	89.4	89.2		
45	113	88.6	88.6	88.6	88.7			
50	122	87.7	87.7	87.8				

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Normal Take-off Thrust Setting, %N₁ 10th Stage Bleeds On –
PACKs On <0005>

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING 10th STAGE BLEEDS ON – PACKS ON CF34–3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	–1000	0	2000	4000	6000	8000	10000
–40	–40	80.9	81.3	82.1	82.9	83.9	84.9	85.9
–35	–31	81.7	82.1	82.9	83.8	84.7	85.7	86.8
–30	–22	82.5	82.9	83.7	84.6	85.6	86.6	87.7
–25	–13	83.3	83.7	84.6	85.5	86.4	87.5	88.5
–20	–4	84.1	84.5	85.4	86.3	87.3	88.3	89.4
–15	5	84.9	85.4	86.2	87.1	88.1	89.2	90.3
–10	14	85.7	86.2	87.0	88.0	89.0	90.1	91.2
–5	23	86.5	87.0	87.8	88.8	89.8	90.9	92.0
0	32	87.3	87.8	88.7	89.6	90.7	91.8	92.6
5	41	88.1	88.6	89.5	90.5	91.5	92.2	92.4
10	50	88.9	89.4	90.3	91.3	91.7	91.9	91.8
15	59	89.7	90.2	91.1	91.4	91.4	91.3	91.3
20	68	90.5	90.8	91.0	91.0	90.9	90.8	90.7
25	77	90.6	90.6	90.6	90.5	90.4	90.2	90.0
30	86	90.1	90.2	90.1	89.9	89.8	89.5	89.2
35	95	89.4	89.5	89.5	89.3	89.0	88.8	88.4
40	104	88.6	88.5	88.6	88.5	88.2		
45	113	87.7	87.6	87.7	87.6			
50	122	86.8	86.8	86.8				

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Normal Take-off Thrust Setting, %N₁ Cowl Anti-ice On <0005>

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING COWL ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	80.9	81.3	82.1	82.9	83.9	84.9	85.9
-35	-31	81.7	82.1	82.9	83.8	84.7	85.7	86.8
-30	-22	82.5	82.9	83.7	84.6	85.6	86.6	87.7
-25	-13	83.3	83.7	84.6	85.4	86.4	87.5	88.5
-20	-4	84.1	84.5	85.4	86.3	87.3	88.3	89.4
-15	5	84.9	85.3	86.2	87.1	88.1	89.2	90.3
-10	14	85.7	86.1	87.0	87.9	89.0	90.0	91.2
-5	23	86.5	87.0	87.8	88.8	89.8	90.9	92.0
0	32	87.3	87.8	88.7	89.6	90.7	91.8	92.6
5	41	88.1	88.6	89.5	90.4	91.5	92.2	92.4
10	50	88.9	89.4	90.3	91.3	91.7	91.9	91.8

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Normal Take-off Thrust Setting, %N₁ Wing and Cowl Anti-ice
On <0005>

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING WING AND COWL ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	80.9	81.3	82.1	82.9	83.9	84.9	85.9
-35	-31	81.7	82.1	82.9	83.8	84.7	85.7	86.8
-30	-22	82.5	82.9	83.7	84.6	85.6	86.6	87.6
-25	-13	83.3	83.7	84.6	85.4	86.4	87.4	88.5
-20	-4	84.1	84.5	85.4	86.3	87.3	88.3	88.7
-15	5	84.9	85.3	86.2	87.1	87.8	88.3	88.7
-10	14	85.7	86.1	86.8	87.2	87.8	88.3	88.7
-5	23	86.2	86.6	86.8	87.2	87.8	88.3	88.7
0	32	86.2	86.6	86.8	87.2	87.8	88.3	88.7
5	41	86.2	86.6	86.8	87.2	87.8	88.3	88.6
10	50	86.2	86.6	86.8	87.2	87.8	87.9	87.9

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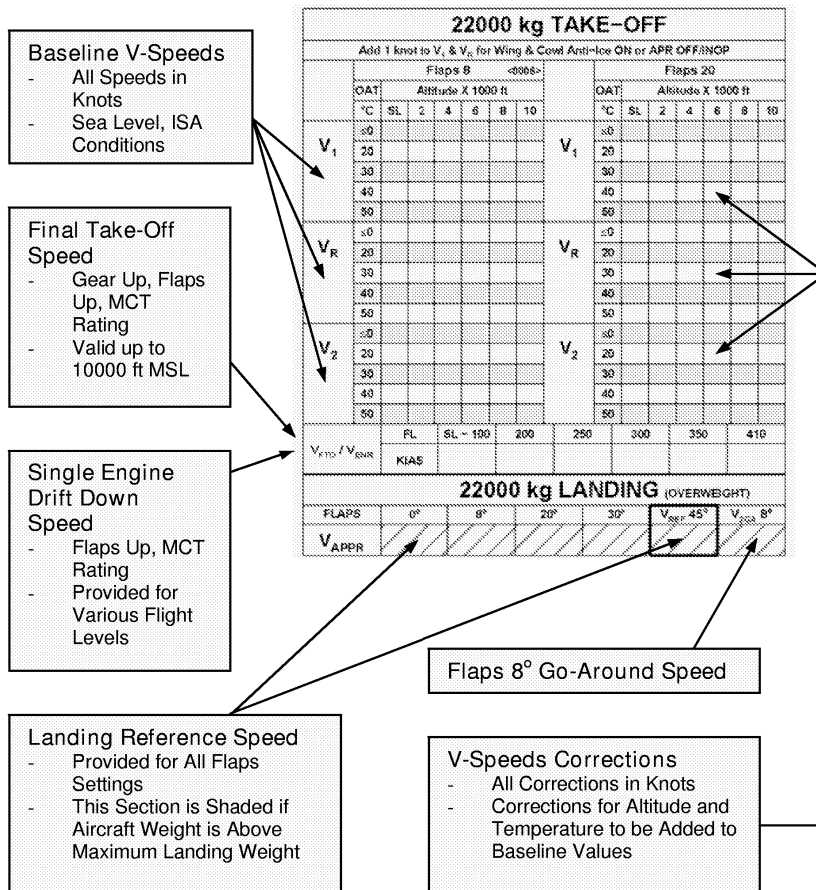
NOTE

In icing conditions, between 0°C and +5°C, at pressure altitudes of 2000 feet and below, the take-off thrust settings with cowl anti-icing on may be used when taking-off with the wing and cowl anti-icing on.



Speedbook

Speedcard – Legend <0006> and <Metric>



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Speedcard – 15400 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

15400 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 108	≤0	0	2	3	5	6	8	V ₁ 98	≤0	0	2	3	5	7	8
	20	0	2	4	6	8	10		20	0	2	4	7	9	11
	30	2	4	6	8	10	12		30	2	4	7	9	11	13
	40	4	6	8	10	12	13		40	4	7	9	11	13	14
	50	6	9	10	12	14	15		50	7	9	11	13	15	17
V _R 113	≤0	0	1	2	3	4	5	V _R 105	≤0	0	1	2	4	5	6
	20	0	1	3	4	5	7		20	0	2	3	5	6	7
	30	1	2	4	5	6	8		30	1	3	4	6	7	8
	40	2	3	5	6	8	9		40	3	4	6	7	8	9
	50	3	5	6	8	9	10		50	4	6	7	8	9	10
V ₂ 127	≤0	0	0	0	0	0	0	V ₂ 117	≤0	0	0	0	0	0	0
	20	0	0	0	0	0	0		20	0	0	0	0	0	0
	30	0	0	0	0	0	0		30	0	0	0	0	0	0
	40	0	0	0	0	0	0		40	0	0	0	0	0	0
	50	0	0	0	0	0	0		50	0	0	0	0	0	0
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		152		152		159		167		175		186	
15400 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		150		138		132		128		120		127			

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Speedcard – 16000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

16000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 112	≤0	-1	1	3	4	6	7	V ₁ 101	≤0	0	2	4	5	7	8
	20	0	1	3	6	8	9		20	0	2	5	7	9	11
	30	1	3	5	7	9	11		30	2	4	7	9	11	13
	40	3	5	8	9	11	12		40	4	7	9	11	13	15
	50	6	8	10	11	13	14		50	7	9	11	13	15	16
V _R 116	≤0	-1	1	2	3	4	5	V _R 108	≤0	0	1	2	3	5	5
	20	0	1	2	4	5	7		20	0	2	3	4	6	7
	30	1	2	4	5	6	8		30	1	3	4	5	7	8
	40	2	3	5	6	8	9		40	3	4	5	7	8	9
	50	3	5	6	7	9	10		50	4	5	7	8	9	9
V ₂ 129	≤0	0	0	0	0	0	0	V ₂ 119	≤0	0	0	0	0	0	0
	20	0	0	0	0	0	0		20	0	0	0	0	0	0
	30	0	0	0	0	0	0		30	0	0	0	0	0	0
	40	0	0	0	0	0	0		40	0	0	0	0	0	0
	50	0	0	0	0	0	0		50	0	0	0	0	0	0
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		154		154		162		170		179		190	
16000 kg LANDING															
FLAPS	0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°				
V _{APPR}	152		140		134		130		122		129				

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